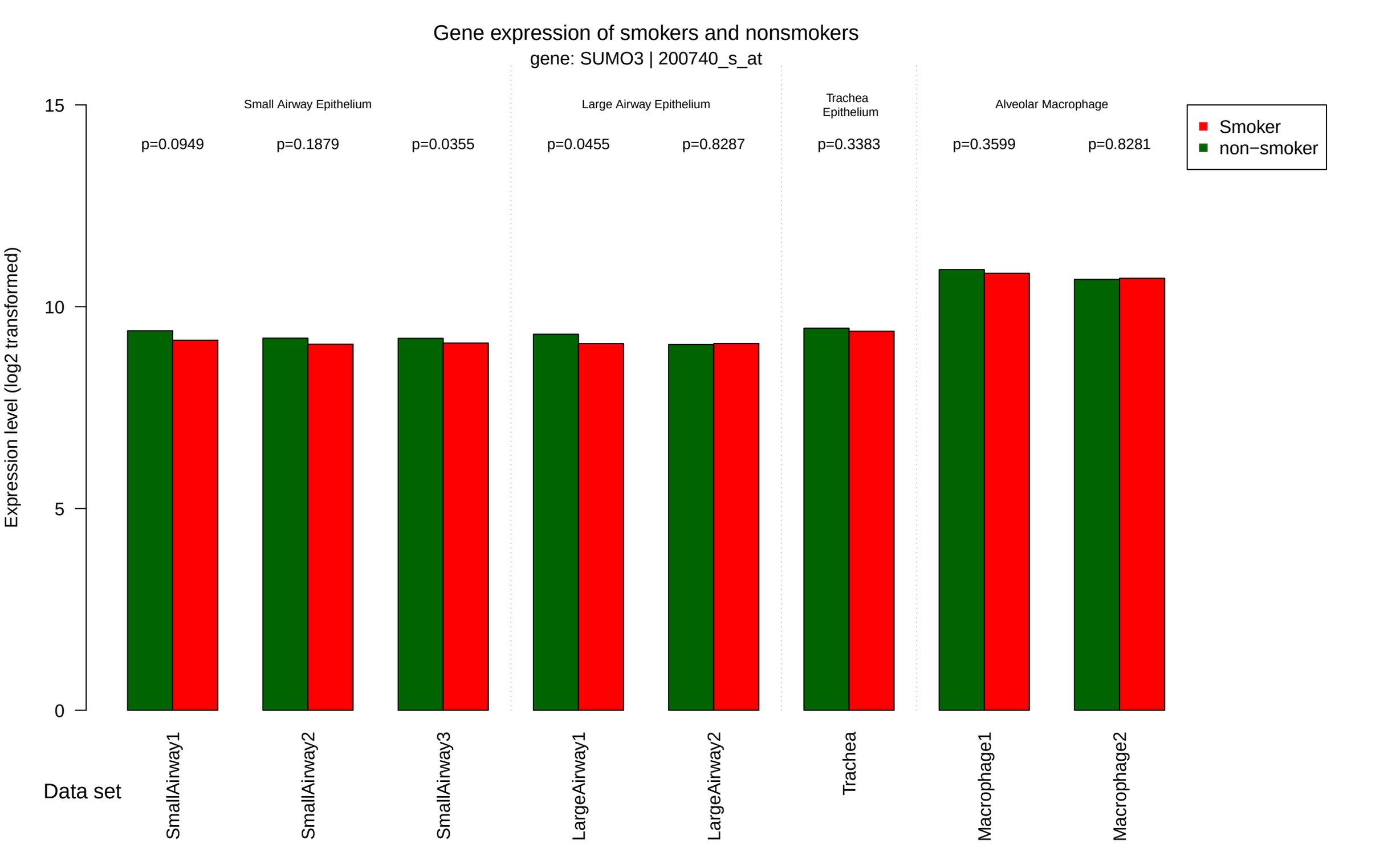
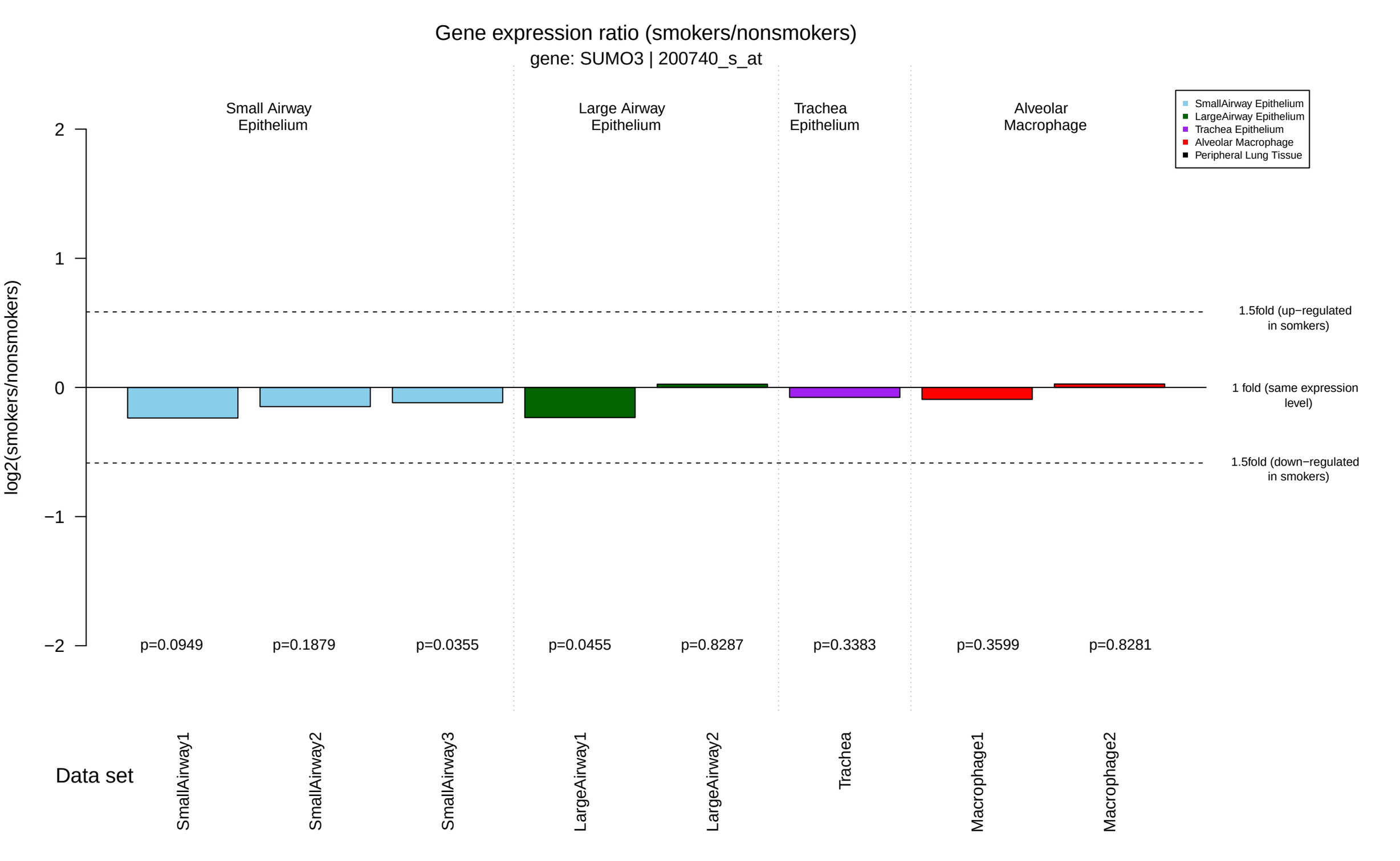
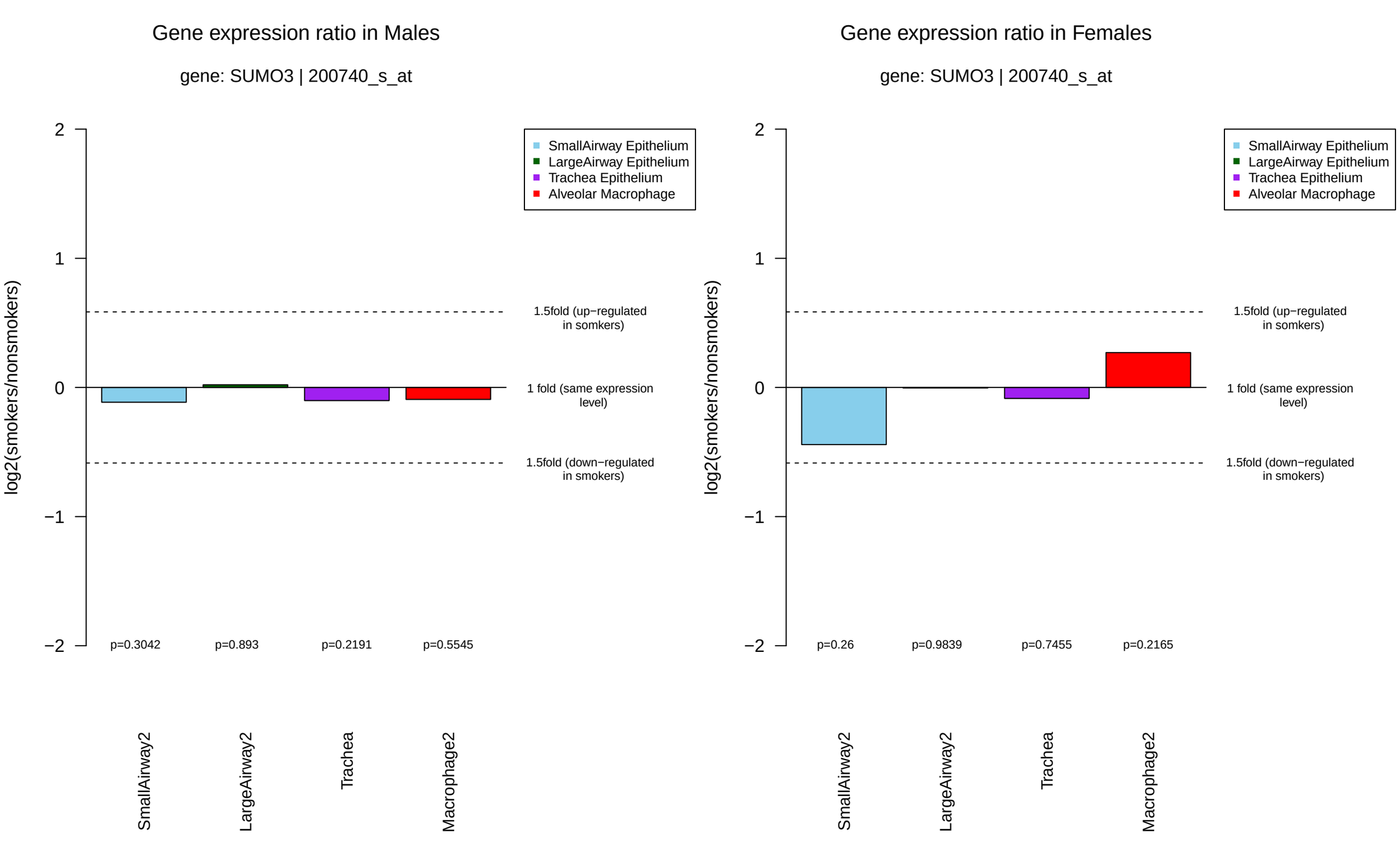
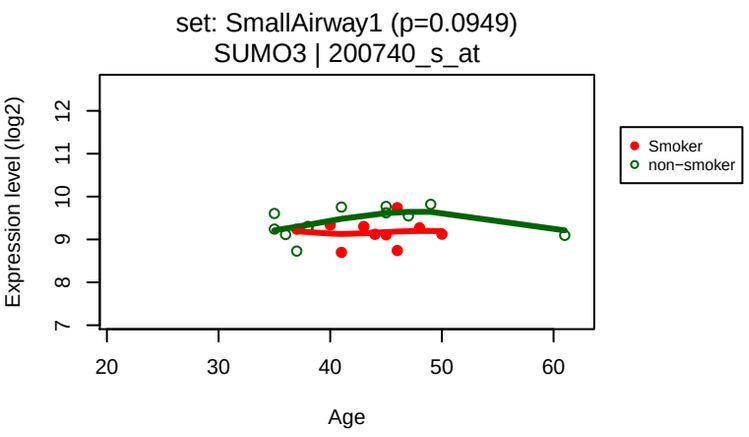
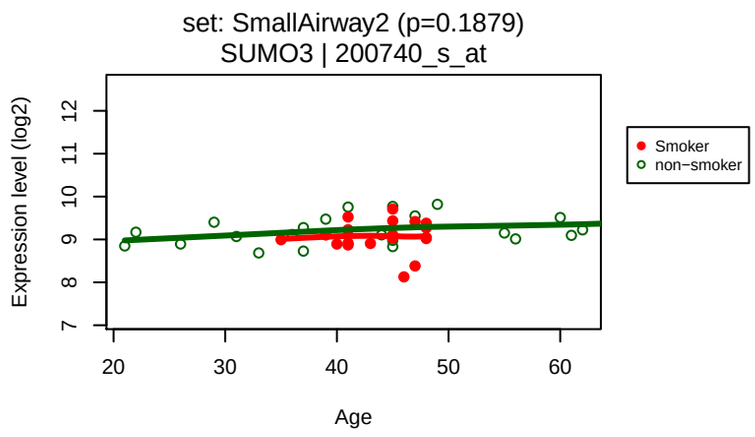
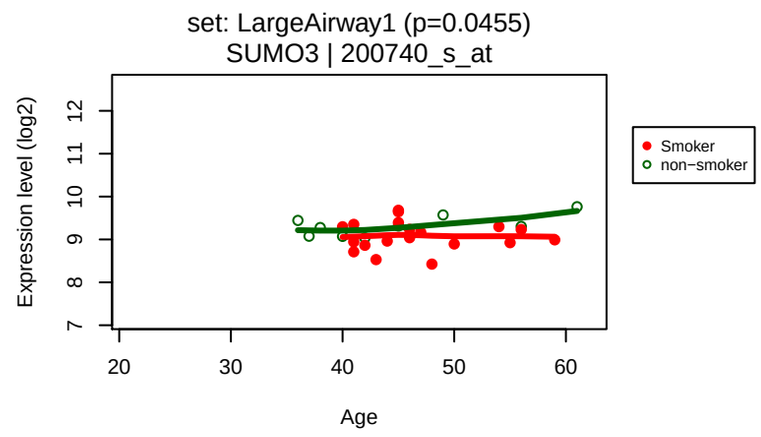

Gene expression of smokers and nonsmokers
gene: SUMO3 | 200740_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0949
p=0.1879
p=0.0355
p=0.0455
p=0.8287
p=0.3383
p=0.3599
p=0.8281
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

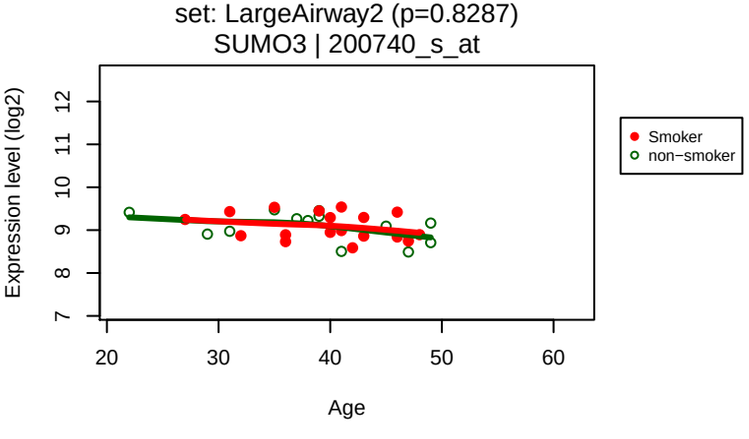

Gene expression ratio (smokers/nonsmokers)
gene: SUMO3 | 200740_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0949
p=0.1879
p=0.0355
p=0.0455
p=0.8287
p=0.3383
p=0.3599
p=0.8281
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

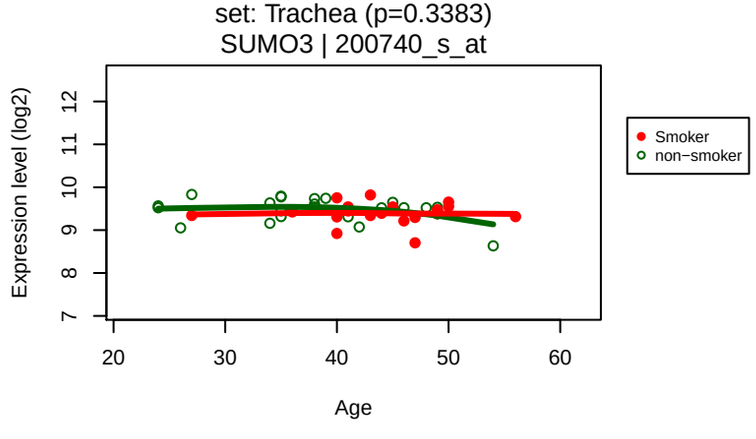

Gene expression ratio in Males
gene: SUMO3 | 200740_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3042
p=0.893
p=0.2191
p=0.5545
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SUMO3 | 200740_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.26
p=0.9839
p=0.7455
p=0.2165
SmallAirway2
LargeAirway2
Trachea
Macrophage2

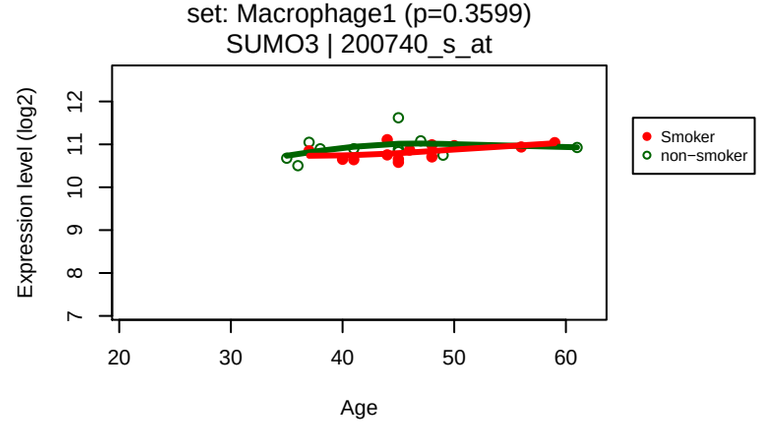

set: SmallAirway1 (p=0.0949)
SUMO3 | 200740_s_at
Expression level (log2)
Age
Smoker
non−smoker

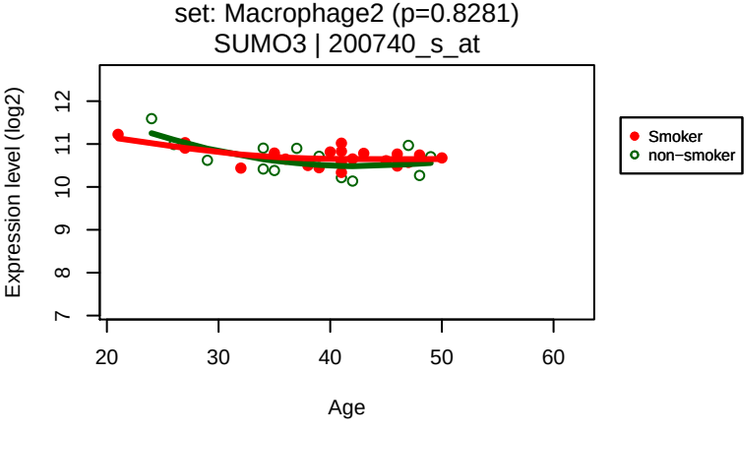

set: SmallAirway2 (p=0.1879)
SUMO3 | 200740_s_at
Expression level (log2)
Age
Smoker
non−smoker

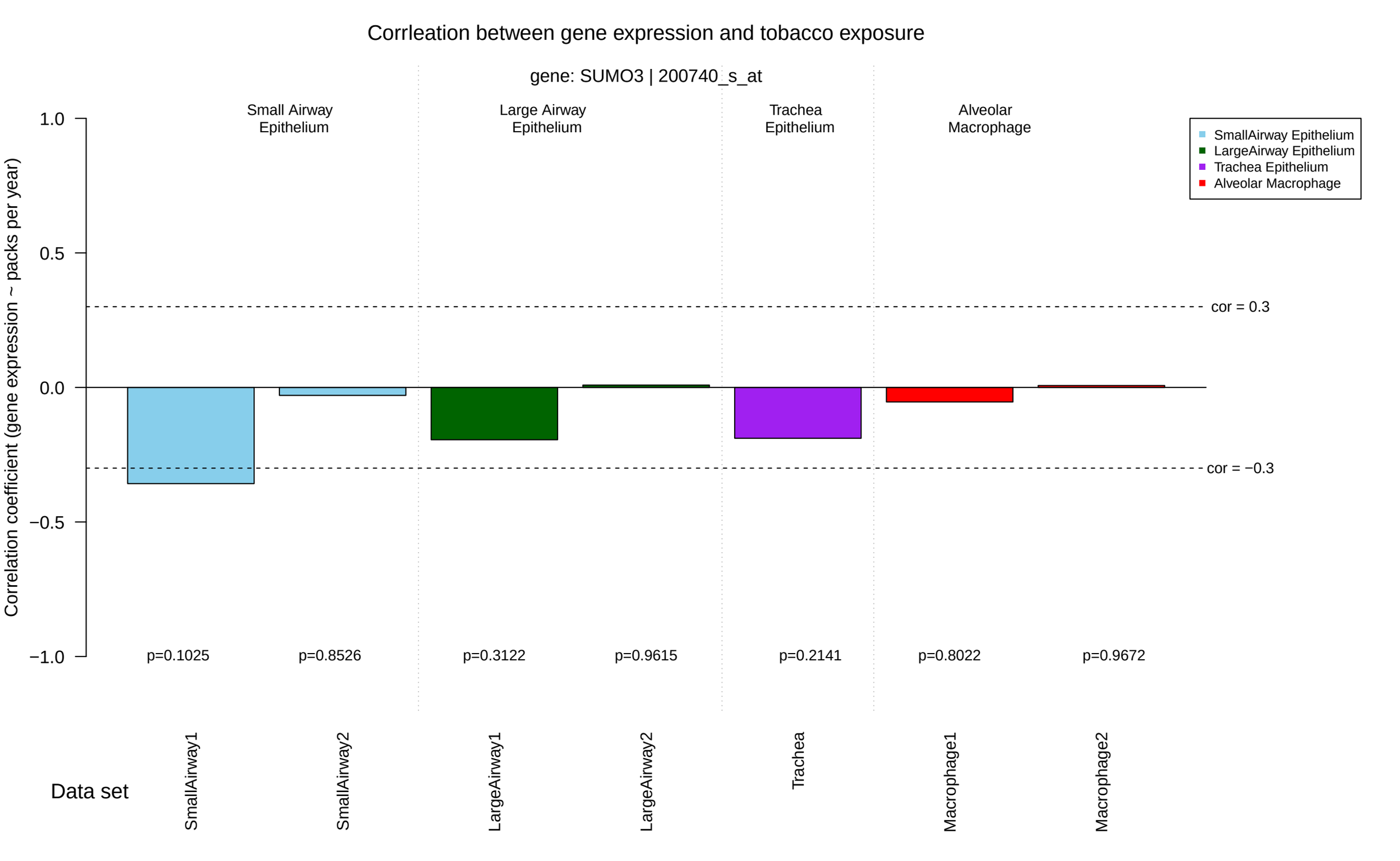

set: LargeAirway1 (p=0.0455)
SUMO3 | 200740_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8287)
SUMO3 | 200740_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3383)
SUMO3 | 200740_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3599)
SUMO3 | 200740_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8281)
SUMO3 | 200740_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SUMO3 | 200740_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1025
p=0.8526
p=0.3122
p=0.9615
p=0.2141
p=0.8022
p=0.9672
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2